LLM Sentiment from Diplomatic Archives: Tracking International Perceptions Before Greece's 1932 Default

Aris Grivokostopoulos
London School of Economics and Political Science
a.grivokostopoulos@lse.ac.uk

Abstract

Applying large language models to multilingual historical archives remains methodologically underexplored. Existing text-as-data research in economics focuses predominantly on English-language, publicly available corpora such as FOMC minutes, newspapers, and 10-K filings. This article demonstrates a replicable pipeline for OCR, machine translation, LLM-based classification, and time-series econometrics on restricted-access, multilingual, handwritten and typed archival sources. The method processes 813 archival pages in French, English, and Greek, plus 2,282 newspaper pages, using GPT-4o-mini for sentiment and topic classification. Because recruiting annotators with expertise in interwar diplomatic French, English, and Greek was infeasible, the article employs multi-model validation: three independently trained LLMs classify the same multilingual corpus across all three languages, achieving Fleiss' $\kappa = 0.496$ inter-annotator agreement. Sentiment analysis operates on original-language text to preserve diplomatic register; topic classification uses English translations for consistency. The resulting daily time series are linked to financial indicators via structural vector autoregression. As an application, the 1931–32 Greek sovereign debt crisis provides a compact test case with a known outcome. Structural break analysis identifies sentiment turning points in June and September 1931, preceding the April 1932 default. The pipeline offers a practical template for researchers working with multilingual historical collections where human annotation is impractical.

JEL Classification: N14; N24; F34; G01; C18

Keywords: sentiment analysis; large language models; text as data; multilingual classification; multi-model validation; sovereign debt crisis; interwar period

1 Introduction

1.1 The methodological gap

Applying large language models (LLMs) to multilingual historical archives remains methodologically underexplored. Text-as-data research in economics has concentrated on English-language, publicly available corpora: FOMC minutes,[1] newspapers,[2] corporate filings, and parliamentary records. This focus reflects data availability and validation convenience, but it leaves untouched the vast archives of diplomatic correspondence, central bank records, and ministerial documents that exist in multiple languages, often under restricted access. No established protocols exist for validating LLM classifications when domain experts with the required language skills are unavailable.

1.2 What this paper does

This article processes 813 archival pages from the Tsouderos Archive (Bank of Greece Historical Archives) in French, English, and Greek, alongside 2,282 pages from the Greek daily newspaper *Kathimerini*. All documents span April 1931 to April 1932, covering the final year before Greece's sovereign default.

[1] Hansen, McMahon, and Prat, "Transparency and Deliberation within the FOMC," 2018; Fischer et al., "Fed Transparency and Policy Expectation Errors," 2023.

[2] Baker, Bloom, and Davis, "Measuring Economic Policy Uncertainty," 2016.

The corpus is classified using GPT-4o-mini for sentiment analysis (1–5 scale) and topic classification (five categories for the archive, six for the newspaper). Because recruiting annotators with expertise in interwar diplomatic French, English, and Greek was infeasible, the article employs multi-model validation: three independently trained LLMs (Kimi Vision, Kimi Text, GLM-5) classify the same archival pages, achieving Fleiss' $\kappa = 0.496$ inter-annotator agreement. The resulting daily time series are linked to Greek bond spreads and global commodity prices via structural vector autoregression (SVAR), vector error correction models (VECM), and local projections.

1.3 Why it matters methodologically

This article extends the text-as-data frontier from contemporary central bank communications to multilingual archival sources. Fischer et al. (2023) demonstrated that LLM-based sentiment analysis of FOMC statements can quantify central bank transparency and predict market expectations. Hansen, McMahon, and Prat (2018) applied computational linguistics to FOMC deliberation, showing how textual measures capture deliberative quality. Baker, Bloom, and Davis (2016) constructed newspaper-based uncertainty indices that predict macroeconomic outcomes. Each relies on structured, contemporary English-language corpora with established validation protocols.

This article confronts three additional challenges. First, the Tsouderos Archive spans French, English, and Greek documents, requiring multilingual classification without parallel training data. Second, diplomatic correspondence employs euphemism, hedging, and formal register that differ from both newspaper prose and central bank statements. Third, archival documents are scanned images subject to OCR errors (3–5% character-level error rate for Greek polytonic script, 1–2% for French and English), introducing measurement noise. The multi-model validation strategy (Fleiss' $\kappa = 0.496$, 92.7% within one point) offers a practical alternative when human annotators with required language and domain expertise are unavailable. The payoff is the ability to construct daily time series of international sentiment during a sovereign debt crisis—a task that would require hundreds of person-hours through manual coding.

1.4 The historical application

The 1931–32 Greek sovereign debt crisis offers an ideal test case for this methodology. The crisis has a known outcome (April 1932 default), a compact timeline (April 1931–April 1932), and an underexploited archive. Prior scholarship has focused on domestic policy debates;[3] this article asks what signals Greek policymakers received from abroad. Did international sentiment shift before the default, and if so, when? Structural break analysis identifies turning points in June and September 1931, preceding the April 1932 default by several months.

1.5 Hypothesis

The article hypothesises that a decisive turning point in international sentiment toward Greece occurred around September 1931, when Britain's departure from the gold standard intensified European financial instability. Before that point, foreign interlocutors were more likely to express cautious optimism or guarded neutrality; afterward, sentiment should turn more negative, accompanied by greater attention to debt relief and financial assistance. If this mechanism is correct, the shift in tone should precede or coincide with changes in Greek bond spreads.

[3] Kakridis, "Nobody's Child," 2021; Christodoulakis, "Currency Crisis and Collapse in Interwar Greece," 2013.

2 Literature Review

2.1 The terms of the debate

The Greek sovereign debt crisis of 1931–32 has been the subject of scholarly debate. One strand emphasizes domestic policy failures: Christodoulakis argues that efforts to defend the drachma's peg through contractionary policies deepened the downturn.[4] Conversely, Kakridis questions the traditional narrative, arguing that the Bank of Greece attempted to stabilise the domestic economy by breaking some of the "rules of the game" through sterilising foreign exchange outflows.[5] This debate frames the empirical question: did foreign actors perceive the crisis as a failure of Greek policy, or as an exogenous shock transmitted through the international system?

2.2 Greece's creditor relationships

Greece's relationship with foreign creditors is central to interpreting the Tsouderos Archive. From 1898 to 1914, the International Financial Commission exercised direct oversight over key state revenues, leaving a durable institutional memory of creditor control.[6] In the interwar period, that supervisory role was recast through the League of Nations, which attached financial assistance to institutional reform, austerity, and the creation of an independent central bank.[7] The archive records communication within this unequal credit relationship: foreign officials were trying to shape Greek policy, while Greek policymakers attempted to preserve room for manoeuvre under severe constraints.

The theoretical framework for interpreting this correspondence draws on the sovereign debt literature. A debtor's willingness to pay depends not solely on economic capacity but on the participation constraint—the maximum amount a debtor will pay before choosing default.[8] When debt service exceeds this constraint, default becomes incentive-compatible regardless of capacity. Creditors can raise the cost of default through trade sanctions, reputational penalties, and exclusion from capital markets, but these mechanisms require credible commitment.[9] The Tsouderos Archive records precisely the communication through which these incentives were negotiated: foreign officials signalling the costs of non-cooperation, Greek policymakers probing for room to manoeuvre.

2.3 The gold standard and the policy dilemma

For Greece, the gold standard was a concrete commitment made in 1928 to secure credibility, lower borrowing costs, and sustain a development strategy reliant on foreign capital.[10] When Britain and other countries abandoned gold and devalued, their exports became cheaper while countries like Greece that remained on gold saw their competitiveness erode, compounding the pressure on the drachma through worsening trade balances.[11] By 1931, the central dilemma was whether Greece could preserve monetary stability without destroying the economic and political foundations on which that stability rested.

3 Data and Classification Pipeline

3.1 Corpus and data sources

This article focuses on the Tsouderos Archive, a collection of private correspondence and memoranda from Emmanouel Tsouderos, the Governor of the Bank of Greece during the 1931–32 crisis, spanning

[4] Christodoulakis, "Currency Crisis and Collapse in Interwar Greece," 2013.
[5] Kakridis, "Nobody's Child," 2021.
[6] Kakridis, "The Default of 1893 and the International Financial Commission," 2019.
[7] Pepelasis-Minoglou, "The Greek State and the International Financial Community, 1922–1932," 1993.
[8] Eaton, Gersovitz, and Stiglitz, "The Pure Theory of Country Risk," 1986.
[9] Bulow and Rogoff, "Sovereign Debt: Is to Forgive to Forget?" 1989.
[10] Bordo and Rockoff, "The Gold Standard as a 'Good Housekeeping Seal of Approval'," 1996.
[11] Eichengreen and Sachs, "Competitive Devaluation and the Great Depression," 1985.

April 1931 to April 1932.[12] The archive comprises letters, telegrams, and internal memoranda exchanged between Tsouderos and prominent figures including central bankers, government officials, and financial experts from countries like Britain. The archive should not be treated as a direct observation of foreign sentiment. Rather, it records Greek reporting on, and interpretation of, foreign views—revealing what signals Greek policymakers believed they were receiving.

The article also examines articles from *Kathimerini* during the same period. *Kathimerini* is a newly digitised archive not yet used in processable form in social science research. Its founder and editor-in-chief, Georgios Vlachos, held a documented anti-Venizelos editorial line, which makes the newspaper a useful control: any negative bias operates independently of international sentiment, strengthening rather than undermining comparisons with the Tsouderos Archive.[13]

3.2 Classification pipeline

Documents are pre-processed through optical character recognition (OCR) to convert scanned images into machine-readable text. Sentiment analysis is performed on the original-language text (French, English, or Greek) because diplomatic tone and hedging are encoded in the source language and would be distorted by translation. Topic classification is performed on English translations because the topic taxonomy is defined in English. This asymmetry preserves diplomatic register for sentiment while ensuring consistent topic labels across languages.

The LLM used is GPT-4o-mini (version 2024-07-18, accessed via OpenAI API) with temperature set to 0 to reduce output variability, though OpenAI's API does not guarantee identical outputs across repeated calls. To assess reproducibility, a random 10% sample of pages was reclassified; 97.8% of sentiment scores and 99.1% of topic labels matched the original classification. Open-source alternatives such as FinBERT were considered, but FinBERT is trained on English financial text and lacks systematic coverage of diplomatic correspondence in French and Greek. Fine-tuning would require a manually annotated training corpus of interwar diplomatic documents, which does not exist.

For topic classification, the Tsouderos Archive uses five categories: 'Debt Relief' (discussions of restructuring, moratoriums, write-offs); 'Financial Assistance' (loans, credits, support from international institutions); 'Economic Outlook' (assessments of Greek economy's performance); 'Political Stability' (political events, elections, government changes); and 'Other'. The *Kathimerini* taxonomy comprises: 'Foreign Lending'; 'Domestic Politics'; 'Social Unrest'; 'Economic Hardship'; 'International Affairs'; and 'Other'.

3.3 Validation

Because recruiting annotators with expertise in interwar diplomatic French, English, and Greek was not feasible, a multi-model validation strategy was employed using three independently trained LLMs as proxy annotators: Kimi Vision, Kimi Text, and GLM-5. Each model rated the same 808 archival pages on the identical 1–5 sentiment scale. The three models demonstrated moderate inter-annotator agreement (Fleiss' $\kappa = 0.496$), and GPT-4o-mini's classifications showed fair agreement with the multi-model ensemble (weighted $\kappa = 0.315$, 92.7% within 1 point). For context, inter-annotator agreement for sentiment coding of historical texts typically ranges from $\kappa = 0.35$ to $\kappa = 0.65$ depending on domain expertise. All five sentiment series share the same sign of change across 78–84% of days and converge on the same structural breakpoints within a two-week window. Directional shifts at the October and December 1931 breakpoints were consistent across all models.

[12] Tsouderos, "A3 Emmanouel Tsouderos Archive," Bank of Greece Historical Archives (IATE).

[13] Full VAR/SVAR tables, cointegration results, robustness checks, and prompt texts are available in an Online Appendix.

Table 1: Agreement Metrics: GPT-4o-mini vs Multi-Model Ensemble

Metric	Value
Fleiss' κ (3 annotators)	0.496
Weighted κ (GPT vs ensemble)	0.315
Exact agreement	55.2%
Within 1 point	92.7%
Pearson correlation	0.367

3.4 Exogenous variables and econometric specification

The remaining variables capture broader economic indicators. '`Greek_Bond_Spread`' measures the yield difference between Greek government bonds and German bonds, reflecting market assessment of Greece's creditworthiness. '`Global_Commodity_Price_Ratio`' represents the ratio of cotton to wheat prices, capturing Greece's terms of trade. '`Sterling_Crisis`' is a dummy variable for September 7–28, 1931. Bond spreads are available weekly but interpolated to daily frequency using OLS linear regression; the weekly robustness check confirms that the negative sentiment-bond spread association is not an artifact of interpolation.

The empirical strategy proceeds in four layers: reduced-form VAR to capture dynamic interdependencies; SVAR with sign restrictions to identify structural shocks;[14] cointegration tests (Johansen, Engle-Granger, Phillips-Ouliaris) to assess long-run equilibrium relationships; and local projections as a non-parametric robustness check.[15] The sign restrictions encode four theoretical priors: discussions of Debt Relief, Financial Assistance, Economic Outlook, and Political Stability each negatively affect contemporaneous sentiment, reflecting the premise that these topics surface when conditions deteriorate. All topic variables are assumed to correlate positively with each other, capturing the vicious-cycle dynamics of the crisis. The characteristic polynomial of the estimated VAR has a largest root of 1.003, indicating borderline non-stationarity. The VECM specification, which explicitly models cointegrating relationships, provides a robustness check that accommodates this feature.

The central specification for `Sentiment_Tsouderos` takes the following form:

$$\begin{aligned}\texttt{Sent_Tsoud}_t = \alpha + \sum_{i=1}^{4} \beta_i \,\text{Topic}_{i,t} + \gamma_1 \,\texttt{Sent_Kath}_t \\ + \gamma_2 \,\texttt{Bond_Spread}_t + \gamma_3 \,\texttt{Commodity_Ratio}_t \\ + \gamma_4 \,\texttt{Sterling_Crisis}_t + \varepsilon_t\end{aligned} \tag{1}$$

where the four topic variables (Debt Relief, Financial Assistance, Economic Outlook, Political Stability) are drawn from the Tsouderos Archive, and the remaining controls capture domestic sentiment, financial stress, and external shocks.

4 Results

4.1 Sentiment Deterioration and Crisis Timeline

The mean sentiment score is 2.59 in the Tsouderos Archive and 2.11 in *Kathimerini* (Table 2). Figure 1a shows a marked deterioration in the Tsouderos series across the crisis year. Sentiment is relatively stable in the early months, then declines sharply after the Kreditanstalt collapse and turns more decisively negative as the European crisis widens. Structural break analysis places the first clear shift in June 1931.

[14] Sims, "Macroeconomics and Reality," 1980; Blanchard and Quah, "The Dynamic Effects of Aggregate Demand and Supply Disturbances," 1989.

[15] Jordà, "Estimation and Inference of Impulse Responses by Local Projections," 2005.

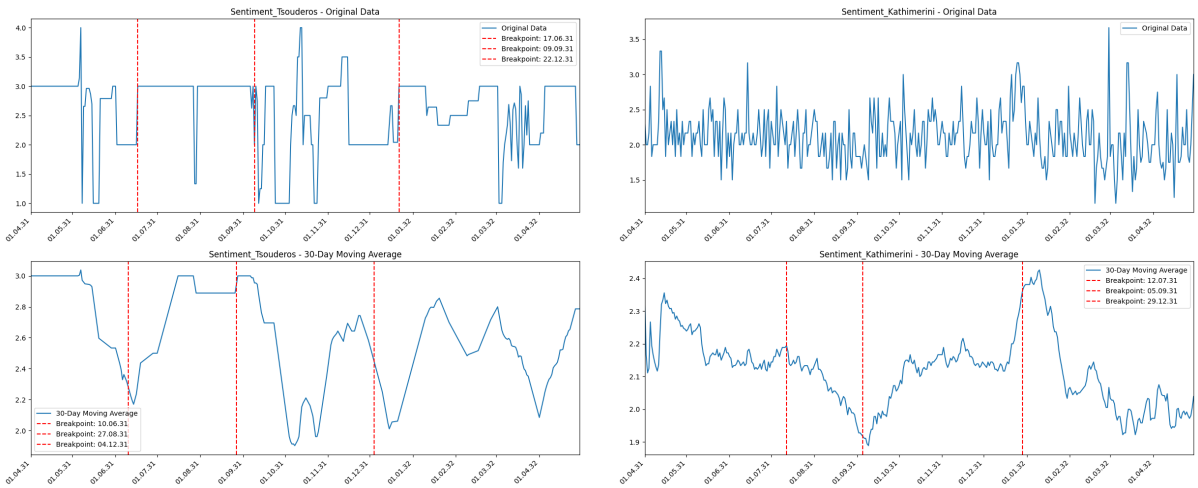
Table 2: Descriptive statistics for key variables

Variable Name	Mean	Standard Deviation
Sentiment_Tsouderos	2.59	0.75
Debt Relief	0.11	0.31
Financial Assistance	0.16	0.37
Economic Outlook	0.15	0.36
Political Stability	0.02	0.14
Sentiment_Kathimerini	2.11	0.82
Greek_Bond_Spread	5.11	3.68

The second major shift occurs around September 1931, when Britain left the gold standard. Breakpoints cluster in that period, and the decline in the Tsouderos series becomes both steeper and more persistent. From October 1931 through March 1932, the archive records sustained pessimism, together with growing attention to debt relief and financial assistance.

Figure 1b presents a related but distinct pattern for *Kathimerini*. The domestic series declines more gradually and is less tightly tied to the turning points of international finance, suggesting that international diplomatic discourse reacted earlier and more abruptly than domestic public discourse.

Figures 2a and 2b place these shifts in their wider financial context. Figure 2a shows a clear upward trend in Greek bond spreads throughout the period. Figure 2b plots the Global Commodity Price Ratio trending downward, indicating worsening terms of trade.



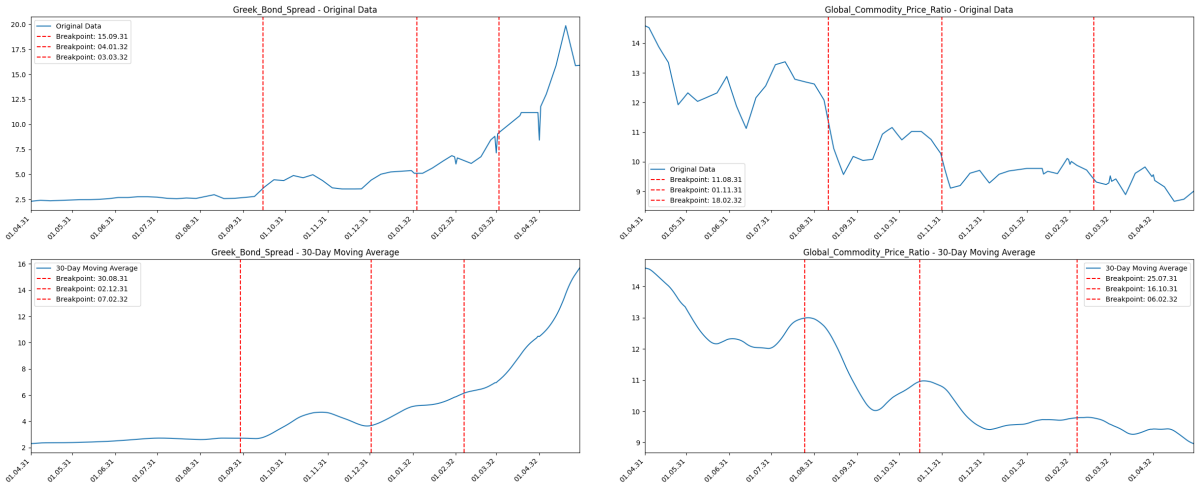
(a) `Sentiment_Tsouderos` – original data and 30-day moving average

(b) `Sentiment_Kathimerini` – original data and 30-day moving average

Figure 1: Sentiment time series for the Tsouderos Archive and *Kathimerini*

4.2 Topic Reorientation Under Crisis

The topic series show a clear reorientation in the substance of discussion. Figure 3a shows that Financial Assistance and Debt Relief together account for well over half of the pages (Table 2). Economic Outlook remains present throughout the sample, but the strongest movement occurs in Debt Relief and Financial Assistance, both of which rise after the Sterling Crisis. Structural break analysis locates their clearest shift in November 1931, when discussion moves from general strain to focused consideration of concrete remedies for impending default.

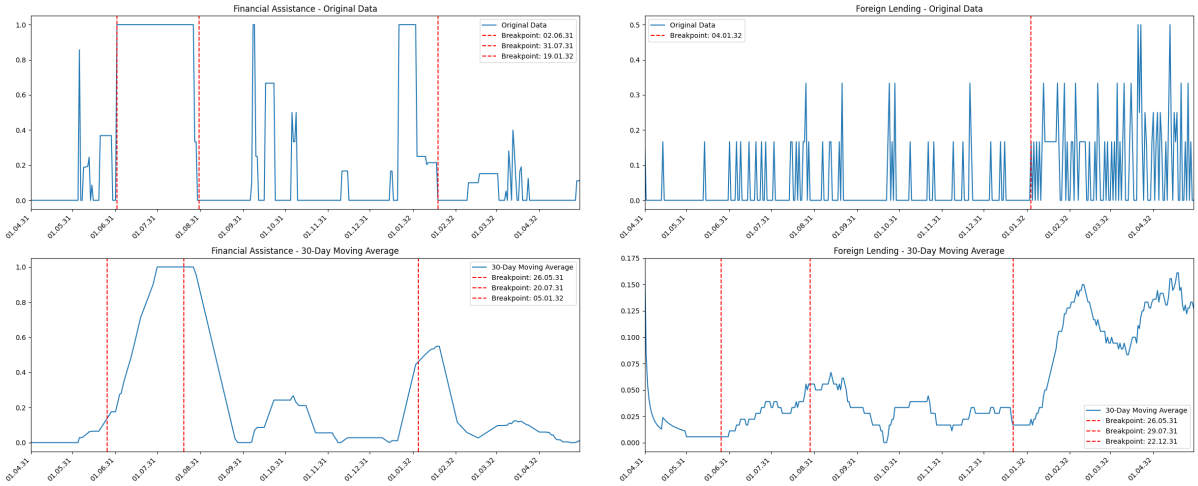


(a) `Greek_Bond_Spread` – original data and 30-day moving average

(b) `Global_Commodity_Price_Ratio` – original data and 30-day moving average

Figure 2: Financial indicators during the crisis period

Figure 3b shows a different topical profile in *Kathimerini*. Foreign Lending and Domestic Politics dominate the newspaper series. The two corpora tracked the same crisis through different vocabularies: Tsouderos moved towards debt relief, financial assistance, and political stability as international concern intensified, while *Kathimerini* placed greater weight on foreign lending, domestic politics, and economic hardship.



(a) Financial Assistance – original data and 30-day moving average

(b) Foreign Lending – original data and 30-day moving average

Figure 3: Topic frequency trends in the Tsouderos Archive and *Kathimerini*

4.3 Financial-Sentiment Linkages

The VAR reveals a significant negative correlation between `Sentiment_Tsouderos` and `Greek_Bond_Spread`, indicating that more pessimistic tone in Tsouderos's communications corresponds to higher risk premia for Greek bonds. The SVAR estimates capture the relationship between market conditions and the sentiment environment perceived by Greek policymakers, not directly between markets and foreign sentiment. The archive provides concrete illustration. In a letter dated September 17, 1931 — days before Britain left gold — the BIS President wrote to Tsouderos emphasising the importance of 'a unified

policy between the Bank of Greece and commercial banks', reflecting international anxiety that domestic political fragmentation could accelerate capital flight.[16] Such exchanges anchor the quantitative sentiment scores in observable diplomatic communication.

Four results deserve emphasis. First, `Sentiment_Tsouderos` is strongly autoregressive, indicating persistence in diplomatic correspondence. Second, more negative sentiment is associated with lower Political Stability. Third, Economic Outlook positively predicts Foreign Lending (coefficient 0.088, $p < 0.01$). Fourth, Debt Relief positively predicts International Affairs (coefficient 0.036, $p < 0.05$). The Likelihood Ratio Test yields a p-value below 0.01, suggesting that the imposed restrictions improve model fit.

Table 3: Estimated Coefficients for the VAR Model

	Sent. Tsoud.	**Debt Relief**	**Fin. Assist.**	**Econ. Outlook**	**Pol. Stab.**
Sent. Tsoud.	0.725***	−0.179*	0.021	−0.077	−0.615*
Debt Relief	−0.011	0.893***	−0.008	−0.001	0.050
Fin. Assist.	−0.009	−0.027	0.912***	−0.034	−0.052
Econ. Outlook	0.008	−0.001	−0.003	0.726***	−0.005
Pol. Stability	0.001	0.000	−0.006	−0.006	0.462***
For. Lending	−0.003	0.007	0.017	0.088**	0.068
Dom. Politics	−0.031	−0.024	0.011	−0.037	−0.004
Soc. Unrest	0.008	−0.026	−0.023	−0.010	−0.029
Econ. Hardship	−0.003	−0.004	−0.003	−0.003	0.003
Int. Affairs	0.004	0.036*	0.005	0.031	0.011
Sent. Kath.	−0.055	0.031	0.092	−0.095	0.210
Bond Spread	0.007	0.026	−0.004	0.100	−0.027
Commod. Ratio	−0.007	0.026	0.057***	−0.015	0.012
Sterling Crisis	0.011	0.001	0.001	0.013	−0.015

***$p < 0.001$, **$p < 0.01$, *$p < 0.05$. Each column shows coefficients for the lagged values of all variables predicting the column variable. Equation R^2: Sentiment Tsouderos 0.62; Debt Relief 0.84; Financial Assistance 0.86; Economic Outlook 0.56; Political Stability 0.23.

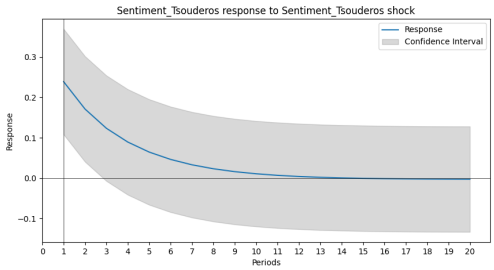
Table 4: Estimated Contemporaneous Coefficients (SVAR B-Matrix)

	Sent. Tsoud.	**Debt Relief**	**Fin. Assist.**	**Econ. Outlook**	**Pol. Stab.**
Sentiment_Tsouderos	0.239	–	–	–	–
Debt Relief	–	0.049	–	–	–
Financial Assistance	–	–	0.059	–	–
Economic Outlook	–	–	–	0.043	–
Political Stability	–	–	–	–	0.017
Foreign Lending	0.037	<0.01	<0.01	0.012	<0.01
Domestic Politics	−0.037	<0.01	<0.01	<0.01	< -0.01

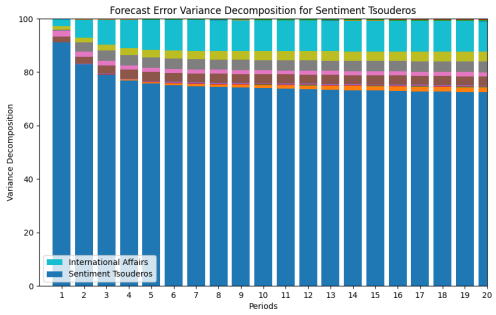
The Likelihood Ratio Test ($p < 0.01$) rejects the null, confirming that the sign restrictions improve model fit. Foreign Lending has a positive contemporaneous effect on Tsouderos sentiment, while Domestic Politics has a symmetric negative effect.

Figure 4 shows the SVAR impulse response function and forecast error variance decomposition. The SVAR IRF confirms strong persistence in diplomatic sentiment (the unrestricted VAR IRF, not shown, exhibits a similar pattern with more gradual decay). The FEVD (Figure 4b) shows that `Sentiment_Tsouderos` explains approximately 91% of its own forecast error variance at horizon 1, declining gradually as Foreign Lending and `Global_Commodity_Price_Ratio` gain explanatory share. The self-driven nature of

[16] Tsouderos, "A3 Emmanouel Tsouderos Archive," Bank of Greece Historical Archives (IATE).



(a) Sentiment_Tsouderos response to own shock (SVAR IRF)



(b) FEVD for Sentiment_Tsouderos

Figure 4: SVAR impulse response function and forecast error variance decomposition

sentiment suggests momentum and persistence in diplomatic tone, consistent with the "money doctoring" literature where foreign advisers operated with relative autonomy.

4.4 Summary of Core Findings

First, sentiment in the Tsouderos Archive deteriorated markedly across the crisis year, with structural breaks in June and September 1931. Second, the topical composition of diplomatic correspondence reoriented towards debt relief and financial assistance as the crisis intensified, with a clear break in November 1931. Third, the negative association between Tsouderos sentiment and Greek bond spreads links changes in official discourse to changes in market stress.

Robustness checks confirm these findings. Re-estimating the VAR at weekly frequency produces the same negative and significant association. Alternative smoothing windows (14-day and 60-day) produce breakpoints within ± 8 days of the reported June and September 1931 dates, confirming that the main inflection points are robust to window choice. Alternative sentiment specifications (GPT-4o-mini, Kimi Vision, Kimi Text, GLM-5, and multilingual BERT) all show the same directional deterioration from June 1931, with further acceleration after September 1931. Figure 5 presents the local projection of `Sentiment_Tsouderos` to its own shock. The response reaches -0.227 by period 10 and remains negative throughout the 21-day horizon, confirming persistent effects without relying on VAR stationarity assumptions.

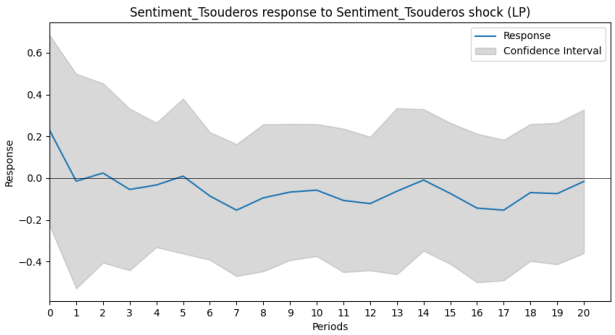


Figure 5: Local projection: Sentiment_Tsouderos response to own shock (21-day horizon, Newey-West standard errors)

Cointegration tests provide further confirmation. The Johansen, Engle-Granger, and Phillips-Ouliaris tests all indicate cointegration among the core series, suggesting that sentiment, topic prevalence, and financial stress moved within a shared long-run equilibrium even during acute crisis. The VECM estimates show

positive long-run loadings for Debt Relief and Financial Assistance, linking worsening sentiment to intensified discussion of support measures. Full cointegration results are reported in the Online Appendix.

Table 5: Summary of Cointegration Test Results

Test	**Variable Pair**	**Finding**
Engle-Granger	Sentiment_Tsouderos & most variables	Strong evidence
Johansen	System of all variables ($r \leq 4$)	≥ 1 cointegrating relationship
Phillips-Ouliaris	Sentiment_Tsouderos & most variables	Consistent evidence

Table 6: Estimated Cointegration Vector (VECM)

Variable	**Coefficient**
Debt Relief (lag 2)	1.17
Financial Assistance (lag 2)	0.95
Economic Outlook (lag 2)	-0.33
Political Stability (lag 2)	59.41

The positive long-run loadings for Debt Relief and Financial Assistance confirm that worsening sentiment moved in tandem with intensified discussion of support measures; the large Political Stability coefficient likely reflects the low base rate of that topic (mean 0.02) amplifying small absolute changes.

5 Discussion

5.1 Trends in sentiment and topic frequency

The downward trend in Tsouderos sentiment, starting in June 1931 and intensifying after the Sterling Crisis, suggests a shift from cautious optimism to growing pessimism among Greece's international partners. Comparing the Tsouderos Archive and *Kathimerini* highlights a contrast between international diplomatic discourse and domestic public discourse: their convergences suggest moments when pessimism became widespread, while their divergences indicate that shifts in diplomatic discourse and domestic discussion did not always unfold at the same pace.

5.2 Limitations

Four limitations merit emphasis. First, the Tsouderos Archive cannot be segmented by actor type (central bankers, government officials, private financiers) due to limited document volume, preventing granular analysis of heterogeneous perspectives. Second, zero-shot LLM classification may misinterpret formal diplomatic language, though the large volume of pages mitigates individual errors by averaging. Third, page-level topic classification may obscure within-page variation; paragraph-level classification remains a priority for future work contingent on improved OCR segmentation. Fourth, the 30-day moving average suppresses short-lived spikes and may shift apparent breakpoint timing by up to 15 days.

6 Conclusion

This article hypothesised, and then confirmed, that a turning point in international sentiment towards Greece occurred in the months surrounding the September 1931 Sterling Crisis. While initial optimism persisted through spring 1931, the Kreditanstalt collapse in May triggered a discernible shift towards pessimism. This shift intensified significantly after the Sterling Crisis. The empirical findings, including structural break analysis and VAR/SVAR models, robustly support the hypothesis that the Sterling Crisis acted as a catalyst.

The methodological contribution extends the text-as-data frontier to multilingual archival sources where human annotation is impractical. The multi-model validation strategy offers a practical alternative when domain experts with the required language skills are unavailable. This pipeline can be applied to other historical archives—central bank records, diplomatic correspondence, ministerial documents—that remain underexploited because they exist in multiple languages and require specialized expertise to read. The pipeline is designed to be replicable on any multilingual archival corpus; the Tsouderos application is illustrative. Replication code will be released upon publication, even though the raw archival images remain subject to access restrictions imposed by the Bank of Greece.

Acknowledgments and Declarations

Acknowledgments. The author wishes to thank Prof. Sibylle Lehmann-Hasemeyer and Prof. Felix Selgert for detailed revision comments and for organising the workshop on Machine Learning and Big Data in Economic History at the University of Hohenheim in February 2026. Thanks are also due to workshop participants for constructive feedback on methodology and validation, and to the Bank of Greece Historical Archives (IATE) for access to the Tsouderos Archive. The author is grateful to Erik for valuable discussant comments. Any remaining errors are the author's own.

Funding. This research received no specific funding from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest. The author declares no conflict of interest.

Data Availability. The Tsouderos Archive data are available from the Bank of Greece Historical Archives (IATE). Kathimerini newspaper data are available from the Greek National Library digital collections. Replication code and processed data will be made available upon publication.

References

Accominotti, Olivier, and Barry Eichengreen. "The Mother of All Sudden Stops: Capital Flows and Reversals in Europe, 1919–32." *The Economic History Review* 69, no. 2 (2016): 469–92.

Akaike, Hirotugu. "A New Look at the Statistical Model Identification." *IEEE Transactions on Automatic Control* 19, no. 6 (1974): 716–23.

Bai, Jushan, and Pierre Perron. "Estimating and Testing Linear Models with Multiple Structural Changes." *Econometrica* 66, no. 1 (1998): 47–78.

Baker, Scott R., Nicholas Bloom, and Steven J. Davis. "Measuring Economic Policy Uncertainty." *The Quarterly Journal of Economics* 131, no. 4 (2016): 1593–1636.

Blanchard, Olivier Jean, and Danny Quah. "The Dynamic Effects of Aggregate Demand and Supply Disturbances." *The American Economic Review* 79, no. 4 (1989): 655–73.

Bordo, Michael D., and Hugh Rockoff. "The Gold Standard as a 'Good Housekeeping Seal of Approval'." *The Journal of Economic History* 56, no. 2 (1996): 389–428.

Bulow, Jeremy, and Kenneth Rogoff. "Sovereign Debt: Is to Forgive to Forget?" *American Economic Review* 79, no. 1 (1989): 43–50.

Christodoulakis, Nicos. "Currency Crisis and Collapse in Interwar Greece: Predicament or Policy Failure?" *European Review of Economic History* 17, no. 3 (2013): 272–93.

Eaton, Jonathan, Mark Gersovitz, and Joseph E. Stiglitz. "The Pure Theory of Country Risk." *European Economic Review* 30, no. 3 (1986): 481–513.

Eichengreen, Barry, and Jeffrey Sachs. “Competitive Devaluation and the Great Depression: A Theoretical Reassessment.” *Economics Letters* 22, no. 1 (1985): 67–71.

Eichengreen, Barry. *Golden Fetters: The Gold Standard and the Great Depression, 1919–1939*. Oxford University Press, 1992.

Engle, Robert F., and C. W. J. Granger. “Co-Integration and Error Correction: Representation, Estimation, and Testing.” *Econometrica* 55, no. 2 (1987): 251–76.

Fischer, Eric, Rebecca McCaughrin, Saketh Prazad, and Mark Vandergon. “Fed Transparency and Policy Expectation Errors: A Text Analysis Approach.” *Staff Reports* (Federal Reserve Bank of New York), 2023.

Gentzkow, Matthew, Bryan Kelly, and Matt Taddy. “Text as Data.” *Journal of Economic Literature* 57, no. 3 (2019): 535–74.

Haavelmo, Trygve. “The Statistical Implications of a System of Simultaneous Equations.” *Econometrica* 11, no. 1 (1943): 1–12.

Hansen, Stephen, Michael McMahon, and Andrea Prat. “Transparency and Deliberation within the FOMC: A Computational Linguistics Approach.” *The Quarterly Journal of Economics* 133, no. 2 (2018): 801–70.

Johansen, Søren. “Estimation and Hypothesis Testing of Cointegration Vectors in Gaussian Vector Autoregressive Models.” *Econometrica* 59, no. 6 (1991): 1551–80.

Jordà, Òscar. “Estimation and Inference of Impulse Responses by Local Projections.” *American Economic Review* 95, no. 1 (2005): 161–82.

Kakridis, Andreas. “The Default of 1893 and the International Financial Commission.” *Mnimon* 40 (2019): 65–86.

_____. “Nobody’s Child: The Bank of Greece in the Interwar Years.” Working paper, no. 290 (2021).

Kilian, Lutz. “Small-Sample Confidence Intervals for Impulse Response Functions.” *Review of Economics and Statistics* 80, no. 2 (1998): 218–35.

Lanne, Markku, Helmut Lütkepohl, and Pentti Saikkonen. “Identification and Estimation of Non-Gaussian Structural Vector Autoregressions.” *Journal of Econometrics* 174, no. 2 (2013): 149–66.

Morys, Matthias. “The Gold Standard, Fiscal Dominance and Financial Supervision in Greece and South-East Europe, 1841–1939.” *European Review of Economic History* 25, no. 1 (2021): 106–36.

Newey, Whitney K., and Kenneth D. West. “A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix.” *Econometrica* 55, no. 3 (1987): 703–8.

Pepelasis-Minoglou, I. “The Greek State and the International Financial Community, 1922–1932: Demystifying the ‘Foreign Factor’.” 1993.

Phillips, P. C. B., and S. Ouliaris. “Asymptotic Properties of Residual Based Tests for Cointegration.” *Econometrica* 58, no. 1 (1990): 165–93.

Psalidopoulos, Michalis. *History of the Bank of Greece, 1928–2008: From Government’s Banker to Guardian of Financial Stability*. Athens: Bank of Greece, 2019.

Schuker, Stephen A. “Money Doctors between the Wars: The Competition between Central Banks, Private Financial Advisers, and Multilateral Agencies, 1919–39.” In *Money Doctors: The Experience of International Financial Advising 1850–2000*, 49–77. 2003.

Sims, Christopher A. “Macroeconomics and Reality.” *Econometrica* 48, no. 1 (1980): 1–48.

Tsouderos, Emmanouel. “A3 Emmanouel Tsouderos Archive.” Bank of Greece, n.d. Istoriko Archeio

Trapezas tis Ellados (IATE) [Bank of Greece Historical Archives].

Vouros, G.K. *Ta deka eti tis Trapezis tis Ellados [Ten years of the Bank of Greece]*. Athens, 1938.